

14. The product $(1\ 3\ 6)(2\ 4)$ of two permutation is :

(A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 6 & 2 & 5 & 1 \end{pmatrix}$.

(B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 1 & 2 & 5 & 4 & 6 \end{pmatrix}$.

(C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 6 & 1 & 4 \end{pmatrix}$.

(D) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 5 & 4 & 6 & 1 \end{pmatrix}$.

15. Which of the following is true ?

- (A) Every function is a permutation if and only if it is one to one.
- (B) The symmetric group S_3 is cyclic.
- (C) The symmetric group S_n is not cyclic for any n .
- (D) Every function from a finite set onto itself must be one to one.

16. Which of the following is an even permutation ?

(A) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 4 & 5 & 1 & 6 & 2 & 1 & 8 \end{pmatrix}$.

(B) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 1 & 4 & 5 & 3 & 7 & 8 & 6 \end{pmatrix}$.

(C) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 4 & 3 & 5 & 2 & 6 & 8 & 7 \end{pmatrix}$.

(D) None of these.

17. What is the largest possible order of a cyclic subgroup of $Z_{12} \times Z_{15}$?

- (A) 60.
- (B) 30.
- (C) 180.
- (D) None of these.

18. In a non-abelian group the element a has order 108. Then the order of a^{12} is :

- (A) 54.
- (B) 27.
- (C) 18.
- (D) 9.

19. f is a homomorphism $f: (R, +) \rightarrow (Z, x)$ such that $f(2) = 3$. Then $f(6)$ is :

- (A) 6.
- (B) 9.
- (C) 18.
- (D) 27.

20. Which of the following is not true ?

- (A) Every subgroup of every group has left cosets.
- (B) A subgroup of a group is a left coset of itself.
- (C) A_n is of index 2 in S_n for $n > 1$.
- (D) None of these.

**FIFTH SEMESTER U.G. DEGREE (SPECIAL) EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 06—ABSTRACT ALGEBRA

Time : Three Hours

Maximum : 120 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. Fill in the blanks : Example of a finite abelian group of order 4 which is not cyclic is _____.
2. Fill in the blanks : Number of units in the ring $\mathbb{Z}_9$ is _____.
3. Fill in the blanks : One way of writing, to show that the identity permutation i is a transposition, is $i =$ _____.
4. Give an example of a group of least order which is not abelian
5. Give an example of a finite integral domain.
6. Compute the inverse of the permutation $\sigma = (1)(1\ 2)(1\ 3)$ in S_4 .
7. Show that the identity element in a group is unique.
8. Find all generators of the group $\mathbb{Z}_5$.
9. Define ring homomorphism.
10. Define permutation.
11. Define division ring.
12. Define a function from $n\mathbb{Z}$ to $\mathbb{Z}$ which is actually an isomorphism.

(12 × 1 = 12 marks)

Turn over

Section B

Answer at least **eight** questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 48.

13. Show that every cyclic group is abelian.
14. Draw the subgroup diagram for $\mathbb{Z}_{12}$.
15. Show that a group is a finite group if it has finite number of subgroups.
16. Let S be a set and let ϕ, ψ and μ be functions mapping S into S . Prove that $\phi * (\psi * \eta) = (\phi * \psi) * \eta$ where the binary operation $*$ is the function composition.
17. Show that intersection of arbitrary number of subgroups is again a subgroup.
18. Compute the product $\sigma\mu$ when $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 4 & 5 & 6 & 2 \end{pmatrix}$ and $\mu = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 5 & 6 & 3 \end{pmatrix}$.
19. Show that any infinite cyclic group is isomorphic to $\mathbb{Z}$.
20. Prove that in any group G , $(a * b)^{-1} = b^{-1} * a^{-1}$ for all elements $a, b \in G$.
21. Show that order of an element in a finite group divides the order of the group.
22. Show that the coset multiplication given by $(Ha)(Hb) = Hab$ is a well defined operation when H is a normal subgroup of G .
23. Find all left cosets of $H = \{0, 3\}$ in $\mathbb{Z}_6$.
24. Calculate $\phi(4)$ if $\phi: \mathbb{Z} \rightarrow \mathbb{Z}$ is a homomorphism that satisfies $\phi(1) = 1$.
25. Is $\mathbb{Q}$, the set of rationals, the field of quotients for integers? Justify your claim.
26. Define binary operation and give an example of it which is not associative and also not commutative.

(8 × 6 = 48 marks)

Section C

*Answer at least **five** questions.*

Each question carries 9 marks.

All questions can be attended.

Overall Ceiling 45.

27. Show that every permutation σ of a finite set is a product of disjoint cycles.
28. Let G be a group. Show that the permutations, $\rho_a : G \rightarrow G$ defined by $\rho_a(x) = ax$, do form a group isomorphic to G .
29. Give any necessary and sufficient condition for a ring R to have no zero divisors, Justify your claim.
30. Define Kernel of a group homomorphism and show that it is a normal subgroup of the domain of the homomorphism.
31. Prove that every field L containing an integral domain D contains the field of quotients of D .
32. Find the index of the subgroup $\langle \sigma \rangle$ in S_5 when $\sigma = (1, 2, 5, 4)(2, 3)$.
33. If the index of a subgroup is 2, show that it is a normal subgroup.
34. If $H \leq G$ and if we define a relation R in G by aRb if $a^{-1}b \in H$, show that it is an equivalence relation?
35. Prove or disprove : (i) Every finite field is an integral domain ; and (ii) Every integral domain is a field.

(5 × 9 = 45 marks)

Section D

*Answer any **one** question.*

The question carries 15 marks.

36. (a) State and prove Lagranges theorem.
- (b) Express $\sigma = (1\ 2\ 3)^{-1}(1\ 3\ 4)^2 \in S_4$ as a product of disjoint cycles.

Turn over

37. (a) If $\phi : G \rightarrow G'$ is a group homomorphism then show that image of a subgroup of G is a subgroup of G' .
- (b) Show that the set of all even permutations form a subgroup of group of Permutations.
38. (a) Give an example of a finite group of order 3 and verify Cayley's theorem for that group.
- (b) Find all the units in the ring $\langle \mathbb{Z}_{12}, +_{12}, \times_{12} \rangle$.

(1 × 15 = 15 marks)

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**FIFTH SEMESTER U.G. DEGREE [SPECIAL] EXAMINATION
NOVEMBER 2020**

(CUCBCSS—UG)

Mathematics

MAT 5B 05—VECTOR CALCULUS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 5B 05—VECTOR CALCULUS
(Multiple Choice Questions for SDE Candidates)

1. The angle between the vectors $a = [1, 2, 3]$ and $b = [0, -2, 1]$ is :

(A) $\cos^{-1} \frac{1}{\sqrt{60}}$.

(C) $\cos^{-1} \frac{1}{\sqrt{70}}$.

(B) $\cos^{-1} \frac{-1}{\sqrt{70}}$.

(D) $\cos^{-1} \frac{1}{\sqrt{80}}$.
2. The straight line through the point $(1, 3)$ in the $x y$ plane and perpendicular to the straight line $x - 2y + 2 = 0$ is :

(A) $3x - y = 2$.

(C) $2x + y = 5$.

(B) $x + y = 1$.

(D) $2x - y = 5$.
3. The parametric equations for the line through $(-3, 2, -3)$ and $(1, -1, 4)$ are :

(A) $x = 1 + 4t, y = -1 - 3t, z = 4 + 7t$.

(C) $x = 3 + 4t, y = 8 - 3t, z = 5 + 7t$.

(B) $x = 2 + 4t, y = -2 - 3t, z = -4 + 7t$.

(D) $x = 1 - 4t, y = -1 + 3t, z = -4 - 7t$.
4. The point of intersection of the line $x = \frac{8}{3} + 2t, y = -2t, z = 1 + t$ and the plane $3x + 2y + 6z = 6$ is :

(A) $(1, 1, 2)$.

(C) $\left(\frac{2}{3}, 2, 0\right)$.

(B) $(2, 0, 1)$.

(D) $(0, 1, 3)$.
5. If $r(t) = \sin ti + e^{-t}j + 3k$, then $\frac{dr}{dt}$ is :

(A) $\sin ti + 3k$.

(C) $\cos ti - e^{-t}j$.

(B) $\cos ti + e^{-t}j + 3k$.

(D) $\sin ti - e^{-t}j$.
6. The domain of the function $f(x, y, z) = xy \ln(z)$:

(A) Entire Space

(C) Half space $z > 0$.

(B) $\{(x, y, z) : xyz \neq 0\}$.

(D) Half space $z < 0$.
7. Which of the following holds for the function $f(x, y) = \frac{4x^6y^2}{x^{12} + y^4}$?

(A) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists.

(C) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

(B) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ doesn't exist.

(D) None of these.

8. Let $f(x, y) = x - y$ and $g(z, y) = e^z$ be two continuous functions. Then the composition function $g(f(x, y)) = e^{x-y}$ is :
- (A) Discontinuous. (B) Continuous.
(C) Continuous at origin. (D) None of these.
9. The function $f(x, y) = xy$ has a :
- (A) Local maximum.
(B) Local minimum.
(C) Both local maximum and minimum.
(D) No local extreme values.
10. The minimum value that the function $f(x, y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ is :
- (A) 2. (B) -2.
(C) 4. (D) -4.
11. The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. The points on the ellipse that lies closest to the origin are :
- (A) $(1, 0, 0)$ and $(0, 0, 1)$. (B) $(0, 1, 0)$ and $(0, 0, 1)$.
(C) $(1, 0, 0)$ and $(0, 1, 0)$. (D) $(1, 0, 0)$ and $(0, 1, 1)$.
12. What is the value of $\iint xy dx dy$ over the first quadrant of the circle $x^2 + y^2 = a^2$?
- (A) $\frac{a^2}{4}$. (B) $\frac{a^2}{8}$.
(C) $\frac{a^4}{4}$. (D) $\frac{a^4}{8}$.
13. A coil spring lies along the helix $r(t) = (\cos 4t) \mathbf{i} + (\sin 4t) \mathbf{j} + t \mathbf{k}$, $0 \leq t \leq 2\pi$. The spring's density, constant, $\delta = 1$. Then the radius of gyration of the spring about the z -axis is :
- (A) 1. (B) 2.
(C) 3. (D) 4.
14. The gradient field of $f(x, y, z) = xyz$ is :
- (A) $yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$. (B) $xy\mathbf{i} + xz\mathbf{j} + yz\mathbf{k}$.
(C) $xz\mathbf{i} + yz\mathbf{j} + xy\mathbf{k}$. (D) None of these.

15. If $\nabla\phi = (y + y^2 + z^2)i + (x + z + 2xy)j + (y + 2xz)k$ and $\phi(1, 1, 1) = 3$, then what is ϕ ?
- (A) $xz + xy + yz^2 - 1$. (B) $xz + yz + xz^2$.
- (C) $xy^2 + xz^2 - 1$. (D) $xy + xy^2 + xz^2 + yz - 1$.
16. Which among the following is the work done in moving a particle once round a circle C in the xy-plane. Given the circle has centre at the origin and radius 3 and the force field is given by $F = (2x - y + z)i + (x + y - z^2)j + (3x - 2y + 4z)k$.
- (A) 8π . (B) 80π .
- (C) 88π . (D) 18π .
17. If $F = (3x^2 + 6y)j - 14yzj + 20xz^2k$, then the value of $\int_C F \cdot dr$ where C is a curve from $(0, 0, 0)$ to $(1, 1, 1)$ with parametric from $x = t, y = t^2, z = t^3$ is :
- (A) 13. (B) 7.
- (C) 5. (D) 11.
18. If $\hat{r}$ is the unit vector in the direction of r and $r = |r|$, then $\text{div}(\hat{r})$ is :
- (A) $\frac{r}{2}$. (B) r .
- (C) zr . (D) $\frac{z}{r}$.
19. Vector product is :
- (A) Commutative. (B) Anticommutative.
- (C) Associative. (D) Not distributive wet vector addition.
20. If F,G are differentiable vector functions and ϕ is a differentiable scalar function. Then :
- (A) $\text{curl}(F \times G) = (\text{grad } \phi) \times F + \phi \text{curl}(F)$.
- (B) $\text{div}(F \times G) = -F \text{curl } G + G \text{curl } F$.
- (C) $\text{div}(F \times G) = (G, \nabla)F - (F, \nabla)G + F \text{div } G - G \text{div } F$.
- (D) $\text{curl}(F \times G) = F \text{curl } G - G \text{curl } F$.

MAT 5B 05 VECTOR CALCULUS

(Multiple Choice Questions for SDE Candidates)

1. The length of the vector a with initial point $p : (3, 2, 5)$ and terminal point $(5, 1, 3)$ is :
- (A) 3. (B) 4.
(C) 5. (D) 6.
2. The straight line through the point $(1, 3)$ in the x, y plane and perpendicular to the straight line $x - 2y + 2 = 0$ is :
- (A) $3x - y = 2$. (B) $x + y = 1$.
(C) $2x + y = 5$. (D) $2x - y = 5$.
3. The volume of the tetrahedron with co-terminal edges representing the vectors $i + j$, $i - j$ and $2k$ is :
- (A) $\frac{2}{3}$. (B) $\frac{3}{2}$.
(C) $\frac{3}{5}$. (D) $\frac{2}{5}$.
4. The parametric equations for the line through $(-3, 2, -3)$ and $(1, -1, 4)$ are :
- (A) $x = 1 + 4t, y = -1 - 3t, z = 4 + 7t$. (B) $x = 2 + 4t, y = -2 - 3t, z = -4 + 7t$.
(C) $x = 3 + 4t, y = 8 - 3t, z = 5 + 7t$. (D) $x = 1 - 4t, y = -1 + 3t, z = -4 - 7t$.
5. The spherical co-ordinate equation for the cone $z = \sqrt{x^2 + y^2}$ is :
- (A) $\Phi = \Pi$. (B) $\Phi = \Pi/4$.
(C) $\Phi = \Pi/2$. (D) None of these.
6. A particle moves along the curve :
- $x = 3t^2, y = t^2 - 2t, z = t^3$ then the acceleration at $t = 1$ is :
- (A) $6i + 2j + 6k$. (B) $6i + 3k$.
(C) $6i + 6k$. (D) $6i + 2j + 3k$.

7. The unit tangent vector at a point t to the curve $r = a \cos ti + a \sin tj$:

- (A) $-\sin ti - \cos tj$. (B) $-\sin ti + \cos tj$.
 (C) $\cos ti - \sin tj$. (D) $-\sin ti + \cos tj$.

8. The domain of the function $f(x, y, z) = xyz \ln(z)$:

- (A) Entire Space. (B) $\{(x, y, z) : xyz \neq 0\}$.
 (C) Half space $z > 0$. (D) Half space $z < 0$.

9. Which of the following holds for the function $f(x, y) = \frac{x+y}{x-y}$?

- (A) f is continuous everywhere.
 (B) f is continuous nowhere.
 (C) f is continuous on $\{(x, y) \in \mathbb{R}^2 : x \neq y\}$.
 (D) f is continuous on $\{(x, y) \in \mathbb{R}^2 : x = y\}$.

10. If $f(x, y) = x \sin xy$, then the value of $\frac{\partial f}{\partial x}$ at $\left(3, \frac{\pi}{6}\right)$ is :

- (A) 0. (B) 1.
 (C) -1. (D) 2.

11. If $f(x, y) = e^{x+y+1}$ then $\frac{\partial f}{\partial y}$ is :

- (A) e^{x+1} . (B) e^y .
 (C) e^{x+y+1} . (D) NOT.

12. Let $w(p, v, \delta, \gamma, g) = PV + \frac{\gamma\delta^2}{2g}$ then the partial derivative W_δ is given by :

(A) $\frac{v}{2g}$.

(B) $p v + \frac{v g}{2g}$.

(C) $\frac{v g}{g}$.

(D) NOT.

13. The derivative of $f(x, y) = x e^y + \cos(xy)$ at the point $(2, 0)$ in the direction $A = 3i - 4j$ is :

(A) 1.

(B) 0.

(C) -1.

(D) 2.

14. The unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$ is :

(A) $i + j = k$.

(B) $\frac{2}{3}i + 2j + 5k$.

(C) $\frac{1}{3}i - j - 7k$.

(D) $\frac{1}{3}i - \frac{2}{3}j - \frac{2}{3}k$.

15. The function $f(x, y) = xy$ has a :

(A) Local maximum.

(B) Local minimum.

(C) Both local maximum and minimum.

(D) No local extreme values.

16. The point $p(x, y, z)$ closest to the origin on the plane $2x + y - z - 5 = 0$ is :

(A) $\left(\frac{5}{6}, \frac{-5}{6}, \frac{5}{6}\right)$.

(B) $\left(\frac{5}{3}, \frac{5}{6}, \frac{5}{3}\right)$.

(C) $\left(\frac{5}{3}, \frac{5}{6}, \frac{-5}{6}\right)$.

(D) $\left(\frac{5}{6}, \frac{5}{3}, \frac{-5}{3}\right)$.

17. The plane $x + y + z = 1$ cuts the cylinder $x^2 + y^2 = 1$ in an ellipse. The points on the Ellipse that lies closest to the origin are :
- (A) $(1, 0, 0)$ and $(0, 0, 1)$. (B) $(0, 1, 0)$ and $(0, 0, 1)$.
(C) $(1, 0, 0)$ and $(0, 1, 0)$. (D) $(1, 0, 0)$ and $(0, 1, 1)$.
18. Which among the following is the value of $\int_0^1 \int_0^1 xy(x-y) dx dy$?
- (A) 4. (B) $\frac{2}{3}$.
(C) $\frac{8}{3}$. (D) $\frac{4}{3}$.
19. The volume enclosed by the co-ordinate planes and the portion of the plane $x + y + z = 1$ in the first octant is :
- (A) $\frac{1}{2}$. (B) $\frac{1}{3}$.
(C) $\frac{1}{6}$. (D) $\frac{1}{4}$.
20. Which among the following is the value of $\int_1^3 \int_{1/x}^1 \int_0^{\sqrt{xy}} xyz dz dy dx$?
- (A) $\frac{1}{8}(26 + \log 27)$. (B) $\frac{1}{8}(27 - \log 26)$.
(C) $\frac{1}{8}(27 + \log 26)$. (D) $\frac{1}{8}(26 - \log 27)$.

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION, NOVEMBER 2021

(CUCBCSS-UG)

Mathematics

MAT 5B 05—VECTOR CALCULUS

Time : Three Hours

Maximum : 120 Marks

Part A*Answer all questions.**Each question carries 1 mark.*

1. Find the domain and range of $z = \sqrt{25 - x^2 - y^2}$.
2. Evaluate $\lim_{(x,y) \rightarrow (1,-1)} \frac{1+x-y}{2-x+y}$.
3. Define gradient of a scalar function.
4. Compute the divergence of $\vec{f} = xy\vec{i} + yz\vec{j} + xz\vec{k}$.
5. Define solenoidal vector.
6. What do you mean by directional derivative.
7. Write the component test for the differential $M(x, y, z) dx + N(x, y, z) dy + P(x, y, z) dz$ to be exact.
8. Find du if $u = \arcsin \frac{x}{y}$.
9. Fill in the blanks : If $\vec{f}$ and $\vec{g}$ are irrotational vector point functions, then $\nabla \cdot (\vec{f} \times \vec{g}) = \dots$
10. State the normal form of Green's theorem in the plane.
11. Fill in the blanks : If $\vec{a}$ is a constant vector and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, then of $\nabla(\vec{r} \cdot \vec{a}) = \dots$
12. State Stoke's theorem.

(12 × 1 = 12 marks)

Part B*Answer any ten questions.**Each question carries 4 marks.*

13. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$.

Turn over

14. Find the vector normal to the surface $\phi(x, y, z) = xyz$ at $(1, -1, 1)$.
15. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at (π, π, π) from $\sin(x + y) + \sin(y + z) + \sin(x + z) = 0$.
16. Prove that $\nabla(r^n) = nr^{n-2}\vec{r}$.
17. Compute the average value of the function $f(x, y, z) = xyz$ over the boundary of the cube $0 \leq x \leq 2, 0 \leq y \leq 2, 0 \leq z \leq 2$.
18. Evaluate $\int_1^2 \int_3^4 \frac{1}{(x+y)^2} dx dy$.
19. Linearize the function $f(x, y, z) = xy + yz + zx$ at $(1, 1, 1)$.
20. Find the directional derivative of $f(x, y, z) = xy$ at $(1, 2)$.
21. Evaluate $\iint_R (xy) dx dy$ where R is the positive quadrant of the circle of radius a centred at the origin.
22. Find the flow of $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ along the portion of the circular helix $x = \cos t, y = \sin t, z = t; 0 \leq t \leq \pi/2$.
23. Test whether $\vec{f} = (yz)\vec{i} + (xz)\vec{j} + (xy)\vec{k}$ is conservative or not.
24. Prove that $\text{div}(\text{curl} \vec{f}) = 0$.
25. Verify whether the differential $(e^x \cos y + yz) dx + (xz - e^x \sin y) dy + (xy + z) dz$ is exact or not.
26. If S is a closed surface enclosing a volume V then prove that $\iint_S \vec{curl} \vec{f} \cdot \hat{n} dS = 0$.

(10 × 4 = 40 marks)

Part C

*Answer any six question.
Each question carries 7 marks.*

27. Using double integrals prove that $\int_0^\infty e^{-(x^2)} dx = \sqrt{\pi}/2$.
28. Evaluate the line integral $\int_C y dx + x dy$ where C is the boundary of the square $x = 0, x = 1, y = 0$ and $y = 1$.

29. Find the work done by the force field $\vec{f} = 3xy\vec{i} - 58\vec{j} + 10x\vec{k}$ along the space curve $C: \vec{r} = (t^2 + 1)\vec{i} + 2t^2\vec{j} + t^3\vec{k}$ where $0 \leq t \leq 2\pi$.
30. Find angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.
31. Evaluate the volume bounded by $y = x^2, x = y^2$ and the planes $z = 0$ and $z = 3$.
32. Evaluate the area enclosed by the region cut from the plane $x + 2y + 2z = 5$ by the cylinder whose walls are $x = y^2$ and $x = 2 - y^2$.
33. Find the Local extreme values of $f(x, y) = x^2 + y^2 + xy + 3x - 3y + 4$.
34. Evaluate the line integral $\int_C \vec{f} \cdot d\vec{r}$ where C is the boundary of the triangle with vertices $(0, 0, 0), (1, 0, 0), (1, 1, 0)$.
35. Show that $\vec{f} = y \sin z \vec{i} + x \sin z \vec{j} + xy \cos z \vec{k}$ is conservative and find its scalar potential.

(6 × 7 = 42 marks)

Part D*Answer any two question.**Each question carries 13 marks.*

36. (a) State Gauss divergence theorem and use it to evaluate the outward flux of $\vec{f} = xy\vec{i} + yz\vec{j} + xz\vec{k}$ through the surface of the cube cut from the first Octant by the planes $x = y = z = 1$.
- (b) Evaluate $\int_{(1,0,0)}^{(0,1,0)} \sin y \cos x dx + \cos y \sin x dy + dz$.
37. Verify Stoke's Theorem for $\vec{f} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ over the rectangular region bounded by $x = 0, x = a, y = 0, y = a$.
38. Verify the Tangential form of Green's theorem in the plane for the vector Field $\vec{f} = (x - y)\vec{i} + x\vec{j}$ over the region bounded by the unit circle $x^2 + y^2 = 1$.

(2 × 13 = 26 marks)